

Box 1: Metaecosystem multitrait biodiversity model

We outline a multitrait-based model to evaluate the effect of trait architecture, i.e., the covariation among biotic, abiotic and migration traits on biodiversity rescue (Figure 1). We explore two trait architectures rooted in empirical and theoretical studies: the modular and the magic trait integration architecture (Endler, 1995; Armbruster and Wege, 2018). The modular scenario takes into account independent traits, i.e. each trait contributes independently to fitness (Figure 1, Modular). The magic trait architecture incorporates the biotic trait as the main contributor to fitness with the abiotic and the migration traits depending on the evolution of the biotic trait (Fig 1, Magic).

Our model considers a landscape with a network of patches connected by migration events and inhabited by resource and consumer populations comprising several individuals each. Biotic, abiotic and migration trait values are initially sampled from normal distributions each with mean and variance for each trait given by $\mathcal{N}(\mu_b, \sigma_b^2)$, $\mathcal{N}(\mu_a, \sigma_a^2)$, and $\mathcal{N}(\mu_m, \sigma_m^2)$, for the biotic, abiotic, and migration trait, respectively. Population dynamics of resource species i is driven by the trait- and temporal-dependent fitness functions following

$$\begin{aligned}\mathcal{W}_R(\mathbf{z}_{\mathbf{B}_i})_x^t &= 1/(1 + \exp(-\gamma(\mathbf{z}_{\mathbf{B}_i x}^t - \hat{\mathbf{z}}_{\mathbf{B}_{\text{forall } j} x}^t)^2)), \\ \mathcal{W}_R(\mathbf{z}_{\mathbf{A}_i})_x^t &= \exp[-\gamma(\mathbf{z}_{\mathbf{A}_i x}^t - \hat{\mathbf{z}}_{\mathbf{A}_i x}^t)^2], \\ \mathcal{W}_R(\mathbf{z}_{\mathbf{M}_i})_x^t &= \exp[-\gamma(\mathbf{z}_{\mathbf{M}_i x}^t - \hat{\theta}_{\mathbf{D} x})^2].\end{aligned}\tag{1}$$

Similarly, the population dynamics for the consumer species is given by

$$\begin{aligned}\mathcal{W}_C(\mathbf{z}_{\mathbf{B}_i})_x^t &= \exp[-\gamma(\mathbf{z}_{\mathbf{B}_i x}^t - \hat{\mathbf{z}}_{\mathbf{B}_{\text{forall } j} x}^t)^2], \\ \mathcal{W}_C(\mathbf{z}_{\mathbf{A}_i})_x^t &= \exp[-\gamma(\mathbf{z}_{\mathbf{A}_i x}^t - \hat{\mathbf{z}}_{\mathbf{A}_i x}^t)^2], \\ \mathcal{W}_C(\mathbf{z}_{\mathbf{M}_i})_x^t &= \exp[-\gamma(\mathbf{z}_{\mathbf{M}_i x}^t - \hat{\theta}_{\mathbf{D} x})^2],\end{aligned}\tag{2}$$

where $\mathbf{z}_{\mathbf{B}_i x}^t$, $\mathbf{z}_{\mathbf{A}_i x}^t$, and $\mathbf{z}_{\mathbf{M}_i x}^t$ represent the biotic, abiotic, and migration trait value of phenotype i at time t and site x for resources and consumer species, R and C , respectively. Equation 1 can be compactly written for resources as

$$\mathcal{W}_R(\mathbf{z}_{\mathbf{BAM}_i})_x^t = \mathcal{E}[-\gamma([\mathbf{z}_{\alpha_i x}^t - \theta_x^*]^T \mathbf{\Omega}_{\mathbf{BAM}}^{-1} [\mathbf{z}_{\alpha_i x}^t - \theta_x^*])],\tag{3}$$

and for consumers as

$$\mathcal{W}_C(\mathbf{z}_{\mathbf{BAM}_i})_x^t = \exp[-\gamma([\mathbf{z}_{\alpha_{ix}}^t - \theta_x^*]^T \boldsymbol{\Omega}_{\mathbf{BAM}}^{-1} [\mathbf{z}_{\alpha_{ix}}^t - \theta_x^*])], \quad (4)$$

where $\mathcal{E}$ in the resources equation represents the fact that the biotic trait value increases its fitness the more it deviates from the mean and common predator type (i.e., negative-frequency dependent selection, equation 1). θ_x^* is the multivariate fitness optimum for the biotic, abiotic and migration trait in site x , with the asterisk referring to being time-dependent in the biotic, but not in the abiotic and migration traits. $\boldsymbol{\Omega}$ is the covariance matrix (Lande, 1980; Melo and Marroig, 2014), and γ determines the interaction sensitivity to deviations from the biotic, abiotic and migration optimum. The interpretation of γ is as follows: it represents the strength of coevolutionary selection between resource and consumer traits (i.e., γ , y-axis in Figure 1), stabilizing environmental selection and stabilizing migration selection for the biotic, abiotic and migration traits, respectively.

If the covariance matrix, $\boldsymbol{\Omega}$, is diagonal, representing our modular scenario (Fig. 1, left), then there is no correlated stabilizing selection (i.e., Modular scenario was run with off-diagonal values equal to zero, $\boldsymbol{\Omega} = [1 \ 0 \ 0; 0 \ 1 \ 0; 0 \ 0 \ 1]$). If the covariance matrix is not diagonal, our magic scenario (Fig. 1, right), then there is correlated stabilizing selection with the biotic trait determining the selection on the other traits (i.e., Magic scenario was run with the following covariance matrix, $\boldsymbol{\Omega} = [1 \ 0.7 \ 0.7; 0.7 \ 1 \ 0; 0.7 \ 0 \ 1]$). The biotic optimum is time-dependent and we define it as the maximum and minimum mean trait distance across all populations in site x for the resource and consumer biotic trait value, respectively (Biotic trait, equation 1 and 2). Migration optimum is given by the mean distance in the landscape (i.e., each site has coordinates from which we calculate the Euclidian distance between each pair of sites). Phenotypes with migration trait value near to this optimum maximize fitness for this trait only for the modular but not for the magic trait scenario. The ecology underpinning this migration optimum takes into account that non migrating or migrating excessively is more risky than the optimal migration. The abiotic optimum is taken initially as the mean of the distribution for each species and it remains the same during the dynamics.

Population dynamics run as follows: at each time step the time-dependent fitness function in site x , for resource species i , $\mathcal{W}_R(\mathbf{z}_{\mathbf{BAM}_i})_x^t$, and consumer species j , $\mathcal{W}_C(\mathbf{z}_{\mathbf{BAM}_j})_x^t$, determines the probabilities to die. Individuals that die are replaced by offspring from the same site x or from migration with probability $\mathbf{m}$. Offspring's phenotypes, $z_o = [z_{ob}, z_{oa}, z_{om}]$, are inherited from the mother plus an environmental deviation, $\mathbf{e}$, taken from a multivariate uniform distribution with range ± 0.001 in all dimensions (i.e., these values are independently sampled for the modular but correlated following the input covariance matrix for the magic trait scenario). We run our model for 500 replicates each

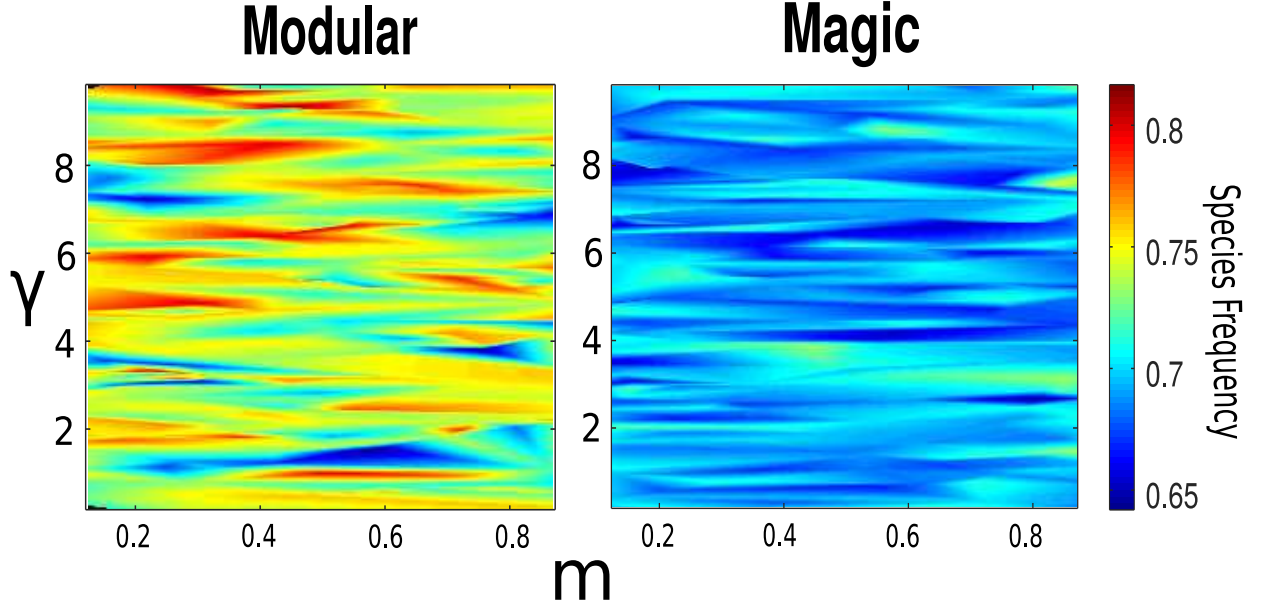


Figure 1: **Effect of modular and magic trait architecture on biodiversity rescue:** We evaluate the importance of coevolutionary selection and migration as biodiversity rescue mechanisms in a Modular (left) and Magic (right), trait architecture scenarios. We explore the formation of hot and cold spots of species persistence in the landscape along a gradient of migration and coevolutionary selection strength. The modular scenario predicts higher biodiversity rescue than the magic one along the migration and coevolutionary selection gradient. Hot spots in biodiversity rescue occur for low-to-medium migration and medium-to-high coevolutionary selection values. The magic scenario predicts a highly homogeneous biodiversity rescue along the gradients

for 50 generations for the two scenarios. To evaluate the importance of coevolution and migration as biodiversity rescue mechanisms, we plot the formation of hot and cold spots of species frequency along a gradient of migration and coevolutionary selection strength. Simulations were run for 10 sites in the landscape each containing 1000 individuals with equal initial abundances for 5 resource and 5 consumer species. Initial mean and variance for the biotic, abiotic, and migration traits for resources and consumers were drawn from normal distributions, $\mathcal{N}(2, 12)$, $\mathcal{N}(50, 9)$, and $\mathcal{N}(60, 2)$. Codes and outputs are available from <https://github.com/melian009/Cometa>.

1. Bibliography

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